When and How to Canonize: a Generalization Perspective

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Abstract

While equivariant architectures are standard for processing symmetric data, there is growing interest in achieving equivariance by applying group averaging or canonization to non-equivariant backbones. However, the theoretical generalization properties of these alternative strategies remain poorly understood. We introduce a theoretical framework to analyze the generalization error of these methods by bounding their covering numbers. We establish a rigorous generalization hierarchy: the error bounds of canonized models are at best equal to the error bounds of structurally equivariant and group-averaged models, and at worst equal to the bounds of non-equivariant baselines. Furthermore, we show that there exist "optimal" canonizations which attain the optimal error bounds, and "poor" canonizations which attain the non-equivariant error bounds, and that this depends on the regularity of the canonization. Finally, applying this framework to permutation groups in point cloud processing, we rigorously prove that the covering number of lexicographical sorting grows exponentially with point cloud dimension, whereas Hilbert curve canonization guarantees polynomial growth. This provides the first formal theoretical justification for the empirical success of Hilbert curve serialization in state-of-the-art point cloud architectures. We conclude with experiments that support our theoretical claims.

1 Introduction

The integration of geometric priors into neural networks has become a cornerstone of modern machine learning, particularly for applications involving graphs, 3D point clouds, and molecular structures. In these domains, the underlying data often exhibits inherent symmetries, meaning the target function to be learned is equivariant to a specific group action. Exploiting these symmetries is known to improve sample complexity and generalization [8, 5].

While specialized architectures, which are equivariant by design, are a common method for processing symmetric data, there is growing interest in achieving equivariance by applying generalized group averaging methods to non-equivariant backbones. This family of methods includes full group averaging, which is theoretically sound but intractable for all but very small groups, group augmentation, which can be seen as an efficient approximation of group averaging, frame averaging [22], which enables equivariant averaging over subsets of the group, and canonization, which maps every group orbit to a single, consistent "canonical" element. Canonization is the most efficient of all the generalized group averaging, as it only involves processing a single element. At the same time, it enjoys the same universal approximation guarantees as full group averaging [14]. These observations have motivated researchers to define canonizations in several diverse domains. These include sign canonizations for spectral embeddings of graphs [19, 18, 12], canonizations of graphs [17] and point clouds [14, 2, 10, 33] including the celebrated Point Transformer V3 [29], and canonization of the SMILES representation of small molecules [28].

In this paper, we provide a theoretical study of canonization from the perspective of generalization. Our focus is both on comparing canonization with other options for enforcing equivariance and on comparing the quality of different canonizations. Our main contributions are threefold:

1. **Canonization vs. Group Averaging:** We prove that the generalization bounds of canonized models are at best equal to the bounds of group-averaged models, and at worst equal to the bounds of non-equivariant models.
2. **Canonizations and continuity:** We show that continuous, isometric canonizations attain the *optimal* generalization error achieved by group averaging, while discontinuous canonizations can lead to *poor* canonizations whose error bounds are identical to non-equivariant models.
3. **Point Clouds Serialization:** We prove that canonizing point clouds with respect to permutation via lexicographical sorting is inferior to the Hilbert canonizations used in Point Transformer V3 [29], thus establishing a theoretical justification for the empirical success of this method.

1.1 Related Work

Canonization vs. Group Averaging It was observed that randomized SMILES can often outperform canonized SMILES [1, 3] (see also [13]). Our first conclusion gives a theoretical justification for this empirical observation.

Discontinuity It was shown in [7, 2] that in many equivariant learning scenarios, continuous canonizations are mathematically impossible. Attempts to mitigate this issue were made in [26, 16]. Our second conclusion complements these works by showing how discontinuity of canonization hurts the generalization of the canonized model.

Equivariance and Generalization Many papers have considered generalization analysis of equivariant models [27, 21, 20, 9], but they do not consider canonizations. To our knowledge, the only paper considering the generalization of canonizations is [25]. They prove that in some setting the *approximation error* of canonized models is lower than group averaged models, and as a result the *expected error* of canonized models can be lower when the model is presented with enough samples. In contrast, our first conclusion is that the *generalization error* of group average models is always lower than the generalization error of canonized models.

1.2 Notation

Throughout the paper, we will consider a metric space (K, ρ), where K is endowed by the action of a group $\mathbb{G}$. We will assume that the metric is $\mathbb{G}$ invariant, i.e., $\rho(g \cdot x, g \cdot y) = \rho(x, y)$ for all $x, y \in K$ and $g \in \mathbb{G}$.

Quotient space We denote the orbit of $x \in K$ by $[x] = \{gx| \quad g \in G\}$, and denote the quotient space to be the space of orbits $K/\mathbb{G} := \{[x]|x \in K\}$. On this space, we define a metric $\rho_{\mathbb{G}}([x], [y]) := min_{g \in \mathbb{G}} \rho(g \cdot x, y)$. The $\mathbb{G}$ invariance of ρ guarantees that $\rho_{\mathbb{G}}$ is symmetric and satisfies the triangle inequality. To ensure that $\rho_{\mathbb{G}}([x], [y]) = 0$ implies that $[x] = [y]$, we explicitly add the assumption that the minimum in the definition of $\rho_{\mathbb{G}}$ always exists. A triplet $(K, \rho, \mathbb{G})$ satisfying the conditions of the last two paragraphs is called a *module*, and all examples considered in this paper are modules.

Covering number Let (K, ρ) be a metric space. We say that $C \subset K$ is an ϵ cover of K, if $\forall x \in K, \exists y \in C : \rho(x, y) \leq \epsilon$. The minimal N such that there exists an ϵ cover of cardinality N is called the ϵ covering number of K, and is denoted by $\mathcal{N}(K, \rho, \epsilon)$.

Canonization We say that $c : K \to K$ is a canonization if for every $x \in K$, (i) x is mapped to an element in its orbit, namely $c(x) \in [x]$, and (ii) all elements in the orbit $[x]$ are mapped to the same element.

2 Generalization, Canonization and Group Averaging

Consider a supervised learning problem of learning an unknown function $f : X \to Y$ from a finite set of n samples $S = \{(x_i, f(x_i)| \quad i = 1, \dots, n)\}$ of the function. Assume that we have some algorithm (e.g., gradient descent) which, given S, returns a hypothesis $h_S : X \to Y$. Let $\ell : Y \times Y \to \mathbb{R}$ be a function which will be used as a loss function. The *expected loss* $\mathcal{L}$ and the *empirical loss* $\mathcal{L}_{\text{emp}}$ are defined via

$$\mathcal{L}_{exp}(h_S) \triangleq \mathbb{E}_{x\sim\mu}\ell(h_S(x), f(x)), \quad \mathcal{L}_{\text{emp}}(h_S) \triangleq \frac{1}{n}\sum_{i=1}^{n}\ell(h_S(x_i), f(x_i))$$

The *generalization* error of h_S is the absolute difference between the expected and empirical loss. The generalization error of h_S can be bounded using the Lipschitz constants of the function h_S, f, ℓ and the covering number of the domain X, as follows:

Theorem 2.1. *Let (X, ρ_X) and (Y, ρ_Y) be metric spaces and assume X is compact. Let $f : X \to Y$ be a c_f Lipschitz function, and let $\ell : Y \times Y \to [0, M]$ be a c_ℓ Lipschitz function. Then for any $\epsilon, \delta > 0$ and natural n, for a training sample set S generated by n IID draws from a distribution μ, with probability of at least $1 - \delta$, we have*

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(X, \rho_X, \epsilon)\ln 2 + 2\ln(1/\delta)}{n}}$$

where c_{h_S} denotes the Lipschitz constant of h_S.

This theorem follows easily from a well-known theorem from [31], which is often employed in the study of neural network generalization [27]. The derivation from the Theorem is given in Appendix A.1.

We now apply this theorem in the setting where the learned function f is invariant to a group action. In the following proposition, we consider three types of models: (i) models that do not exploit function symmetry at all, (ii) models that are invariant by construction, and (iii) non-invariant models composed with canonizations.

Proposition 2.2. *Let $(K, \rho, \mathbb{G})$ be a module. Let (Y, ρ_Y) be a metric space, and assume K is compact. Let $f : K \to Y$ be a c_f Lipschitz function **which is $\mathbb{G}$ invariant**, and let $\ell : Y \times Y \to [0, M]$ be a c_ℓ Lipschitz function , then for any $\epsilon, \delta > 0$ and natural n, for a training sample set S generated by n IID draws from the distribution μ, with probability of at least $1 - \delta$, we have*

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(h_S)\ln 2 + 2\ln(1/\delta)}{n}}$$

where

$$\mathcal{N}(h_S) = \begin{cases} \mathcal{N}(K, \rho, \epsilon) & \text{if } h_s \text{ is } c_{h_s} \text{ Lipschitz} \\ \mathcal{N}(c(K), \rho, \epsilon) & \text{if } h_s = \tilde{h}_s \circ c, \tilde{h}_s \text{ is } c_{h_s} \text{ Lipschitz, and } c \text{ is a canon.} \\ \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) & \text{if } h_s \text{ is } c_{h_s} \text{ Lipschitz and } \mathbb{G} \text{ invariant} \end{cases} \tag{1}$$

Proof idea. The claim for a Lipschitz function h_s without any invariant structure follows immediately from Theorem 2.1. For a function of the form $h_s = \tilde{h}_s \circ c$, the claim follows from Theorem 2.1 when replacing the domain K with $c(K)$ and the function h_s with $\tilde{h}_s$. The claim for h_s which is both Lipschitz and invariant follows from the fact that due to invariance h_s, f can be identified with function $\hat{h}_s, \hat{f} : K/\mathbb{G} \to Y$ satisfying $\hat{h}_s([x]) = h_s(x), \hat{f}([x]) = f(x), \forall x \in K$, and the Lipschitz constant remains unchanged after the identification (see Lemma 20 in [23]). □

For each of the three model types considered in Proposition 2.2, a different value of $\mathcal{N}(h_S)$ is provided. The following simple lemma establishes the relationship between these three values:

to self: add proof for equivariant setting in appendix, mention in main text

Lemma 2.3. *For any module $(K, \rho, \mathbb{G})$, and any $\epsilon > 0$, the following inequality holds:*

$$\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \leq \mathcal{N}(c(K), \rho, \epsilon) \leq \mathcal{N}(K, \rho, \epsilon) \tag{2}$$

Proof. The inequality $\mathcal{N}(c(K), \rho, \epsilon) \leq \mathcal{N}(K, \rho, \epsilon)$ is immediate since $c(K) \subseteq K$. For the other inequality, note that the quotient mapping $q : c(K) \to K/\mathbb{G}$ defined by $q(x) = [x]$ is onto and 1-Lipschitz, and therefore it transforms any ϵ cover of $c(K)$ to an ϵ cover of $K/\mathbb{G}$, thus implying that $\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \leq \mathcal{N}(c(K), \rho, \epsilon)$.

□

Informally, the conclusion we draw from this lemma is as follows:

Conclusion 1

The generalization bounds of canonized models are at best equal to the bounds of group-averaged models, and at worst equal to the bounds of non-equivariant models.

isnt the proof done for invariant models? In general, it feels "equivariant" and "invariant" are switched in and out, which is a bit confusing. Maybe clarify?

While this conclusion is valid for all equivariant models, it is most relevant for equivariant models obtained from group averaging a non-equivariant model. In this setting, the same backbone model can be used for canonization and group averaging, and so a direct comparison can be made.

We conclude this section by showing that the ratio between the largest and smallest covering numbers in (2) is bounded by the group's cardinality:

Proposition 2.4. *[See proof in A.3] Let $(K, \rho, \mathbb{G})$ be a module, and assume $\mathbb{G}$ is finite. Then*

$$\mathcal{N}(K, \rho, \epsilon) \leq |\mathbb{G}| \cdot \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon).$$

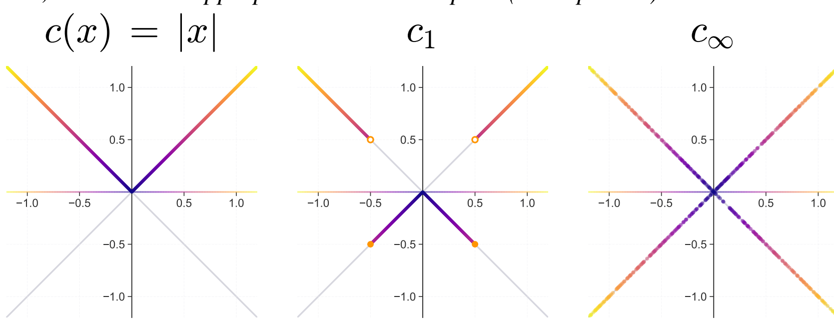


Figure 1: *Visualization of three canonizations for the action of $\mathbb{G} = \{-1, 1\}$ on $\mathbb{R}$. The natural canonization $c(x) = |x|$, which is shown to be optimal (Example 3.3), and two discontinuous canonizations, which on an appropriate domain are poor (Example 3.5).*

3 Canonizations and Continuity

In the previous section, we established upper and lower bounds for the covering numbers of canonizations. Our focus in this section is to study when these bounds are achieved. As we will see, this will be related to the continuity of the canonization. We begin by defining optimal and poor canonizations.

Definition 3.1. Let $(K, \rho, \mathbb{G})$ be a module and let ϵ be a positive number. We say that c is (K, ϵ) *optimal* if it attains the lower bound in (2), namely $\mathcal{N}(c(K), \rho, \epsilon) = \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon)$. We say that c is (K, ϵ) *poor* if it attains the upper bound in (2), namely $\mathcal{N}(c(K), \rho, \epsilon) = \mathcal{N}(K, \rho, \epsilon)$.

We will next show conditions for optimal and poor canonizations, and show how these conditions relate to the continuity of the canonization.

Optimal canonizations: First, we prove canonizations, which are isometries, are optimal. We say that a canonization $c : K \to K$ is an *isometry* if $\rho(c(x), c(y)) = \rho_{\mathbb{G}}([x], [y])$ for all $x, y \in K$.

Claim 3.2. *[proof in A.3] Any isometric canonization $c : K \to K$ is (K, ϵ) optimal for all $\epsilon > 0$.*

We next use this claim to give several examples of optimal canonizations:

Example 3.3. The following canonizations are isometries (see proof in A.3)

1. The canon. $c(x) = |x|$ with respect to the action of $\{-1, 1\}$ on $\mathbb{R}$ and the standard metric.
2. The canon. $\mathrm{sort} : \mathbb{R}^n \to \mathbb{R}^n$ with respect to the action of permutations and a metric ρ induced by a p norm.
3. The centralization canon. defined for point clouds in $\mathbb{R}^{d\times n}$ with respect to the action of translation by vectors in $\mathbb{R}^d$, with the metric ρ induced by the Frobenius norm.

Poor canonizations: We next show that discontinuities in canonizations can lead to poor canonizations. This occurs under two conditions: firstly, the canonization is 'strongly discontinuous' in the sense that it has $|G|$ partial limits. Secondly, the domain K of the group action is chosen adversarially to target the canonization's discontinuity.

Proposition 3.4. *[proof in A.3] Let $(V, \rho, \mathbb{G})$ be a module, where $\mathbb{G}$ is a finite group. Let c be a canonization such that c has $|G|$ partial limits at a point $x \in V$, then for any small enough $\epsilon > 0$ there exists a $\mathbb{G}$ invariant set $K \subseteq V$ such that $\mathcal{N}(c(K), \rho, \epsilon) = \mathcal{N}(K, \rho, \epsilon) = |\mathbb{G}| \cdot \mathcal{N}(K/\mathbb{G}, \rho, \epsilon)$.*

Based on this proposition, we give two examples of poor canonizations:

Example 3.5. Consider the action of $\mathbb{G} = \{-1, 1\}$ on $V = \mathbb{R}$. Consider the canonizations

$$c_1(t) = \begin{cases} |t| & \text{if } |t| > 1/2 \\ -|t| & \text{if } |t| \leq 1/2 \end{cases}, \quad c_\infty(t) = \begin{cases} |t| & \text{if } t \in \mathbb{Q} \\ -|t| & \text{if } t \notin \mathbb{Q} \end{cases}.$$

Both of these canonizations are discontinuous and have a point with $|G| = 2$ partial limits. Therefore, by Proposition 3.4 for every small enough $\epsilon > 0$ we can choose an adversarial domain for which the canonization is poor. For c_1 we can choose for every $\epsilon \in (0, 1)$ the set $K = B_\epsilon(-\frac{1}{2}) \cup B_\epsilon(\frac{1}{2})$ and then $\mathcal{N}(K, \rho, \epsilon) = \mathcal{N}(c_1(K), \rho, \epsilon) = 2$ and so the canonization is poor. In the canonization c_∞ the discontinuities are much more extreme. In this case, we can choose a non-adversarial domain $K = [-1, 1]$, and the canonization will be poor for any $\epsilon > 0$ because the covering number of a set and its closure are the same, and in our case, the closure of $c_\infty(K)$ will be all of K.

Example 3.6. Consider the action of the permutation group $\mathbb{G} = S_n$ on the space of matrices $\mathbb{R}^{d\times n}$ by permuting the columns of the matrices. We call such matrices 'point clouds'. We assume that $n, d \geq 2$ (the case $d = 1$ sorting is an isometry by Example 3.3). A natural canonization in this scenario is c_{lex}, which permutes the columns of the matrices so that the first row of the matrix is sorted from small to large, and in the event of ties in the first row, sorts according to the second row, and continues in this fashion until a permutation is uniquely defined. The canonization c_{lex} is discontinuous. Moreover, the canonization has $n! = |\mathbb{G}|$ partial limits at any matrix $M \in \mathbb{R}^{d\times n}$ satisfying that the matrix does not contain two identical columns, but $M_{1,1} = M_{1,2} = \ldots = M_{1,n}$. Thus, according to Proposition 3.4, for every small enough $\epsilon > 0$ there exists an adversarial domain for which the canonization is poor.

Conclusion 2

We show that isometric canonizations are optimal, while discontinuous canonizations can yield poor canonizations. These results give a rigorous justification for the importance of continuous canonizations.

4 Case study: Permutation Canonizations

In Example 3.6, we showed that lexicographical sorting is a poor canonization on an adversarial domain. In this section, we analyze the covering numbers of lexicographical sorting when considering a fixed, 'natural' domain of point clouds. We will show that even in this non-adversarial setting, the covering number of lexsort canonization is much higher than the covering number of the quotient space: we show this by showing that the covering number of lexsort grows exponentially in n, the number of points in the point cloud, while the covering number of the quotient space only grows polynomially in n. Moreover, we will consider an alternative canonization based on Hilbert curves, and show that its covering numbers also grow only polynomially in n. The significance of this result is that it gives a theoretical justification for the choice of Point Transformer V3 [29] to use Hilbert canonizations for point cloud serialization.

Throughout this section we consider the module $(K, \rho, \mathbb{G})$ with the domain $K = [0,1]^{d\times n}$, the distance induced by the element-wise infinity norm $\rho(X,Y) = \|X - Y\|_\infty$, and the permutation group $\mathbb{G} = S_n$. We will show that for fixed ϵ, d, the covering number of c_{lex} grows exponentially in n, while the covering number of the quotient space and Hilbert canonizations only grows polynomially in n. We begin by proving the exponential growth of lexsort canonizations:

Lemma 4.1. *[proof in A.4] Let $k, d, n \geq 2$ be natural numbers, and set $\epsilon = 1/(2k)$, $K = [0,1]^{d\times n}$ and $\mathbb{G} = S_n$. Then the covering number of the lexicographical sorting canonization satisfies*

$$\left(\frac{1}{2\epsilon}\right)^{(d-1)\cdot n+1} \leq \mathcal{N}(c_{\text{lex}}(K), \rho, \epsilon). \tag{3}$$

Proof idea. The lexsort canonization is ineffective on the set of its 'severe discontinuities'- point clouds whose first row is constant. These point clouds form a $(d-1)\cdot n+1$ dimensional hypercube. The covering number of $c_{\text{lex}}(K)$ is upper bounded by the covering number of this hypercube, which is the left-hand side of (3). $\square$

Next, we prove the polynomial growth of the quotient space's covering number:

Lemma 4.2. *Let n, d be natural numbers and $\epsilon \in (0,1)$. Set $K = [0,1]^{d\times n}$ and $\mathbb{G} = S_n$. Then the covering number of the quotient space satisfies*

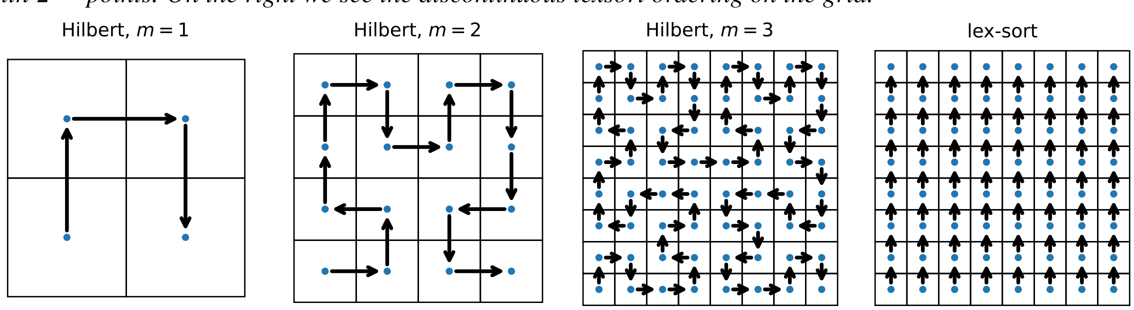
$$\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \leq \binom{n + k^d - 1}{n}, \text{ where } k = \left\lceil \frac{1}{2\epsilon} \right\rceil. \tag{4}$$

Proof. Set $k = \lceil 1/(2\epsilon) \rceil$. The set K can be covered by k^d hypercubes of diameter k and radius $1/(2k) \leq \epsilon$. Let C denote the collection of centers of these hypercubes, and after applying the quotient map $\pi(c) = [c]$ we obtain points $\pi(C)$ in the quotient space which are an ϵ cover of $K/\mathbb{G}$. Note that points in C which are permutations of each other are mapped by π to the same orbit. Therefore, the cardinality of the cover $\pi(C)$ is the number of equivalence classes with respect to the relation on C defined by the group: $c_1 \equiv c_2 \iff \exists \pi \in S_n : c_1 = \pi(c_2)$. As explained in more detail in the proof, the number of equivalence classes is the combinatorial expression in (4). $\square$

4.1 Hilbert Curve Canonization

Finally, we will give an intuitive explanation of the Hilbert canonization and formulate the claim that its covering number exhibits only polynomial growth in the point cloud dimensionality. For a full definition of the canonization and proof see Appendix A.5.

Figure 2: *The Hilbert curves H_m for $d = 2$ and $m = 1, 2, 3$ induce an ordered on a grid in $[0,1]^d$ with 2^{md} points. On the right we see the discontinuous lexsort ordering on the grid.*



The Hilbert curve is a space-filling curve, mapping the unit interval $[0,1]$ continuously onto a hypercube $[0,1]^d$. It is defined as a limit of curves $H_m : [0,1] \to [0,1]^d$ which are not onto but whose image becomes more and more 'dense' in $[0,1]^d$ as m grows, as shown in Figure 2 for the case where $d = 2$. The curves H_m can be used to define an order on $[0,1]^d$ (at least on points in the image of the curve) and thus induce a canonization. Intuitively, the advantage of this order over the lexsort ordering is that this ordering is continuous (see Figure 2) again. However, while previous work [29, 6] cited this continuity as justification for Hilbert ordering, a formal justification of why this continuity is important for learning has not yet been provided. We address this caveat by showing the polynomial growth of the covering number of Hilbert canonizations:

Theorem 4.3. *Let $d, n, m \geq 2$ be natural numbers, let $\epsilon \in (0, 1)$, and assume that $\epsilon > 2^{-m-1}$ set $K = [0, 1]^{d\times n}$ and $\mathbb{G} = S_n$. Then the covering number of the Hilbert canonization c_m satisfies*

$$\mathcal{N}(c_m(K), \rho_\infty, \epsilon) \leq \binom{n + \lceil 1/(2\delta)\rceil - 1}{n}, \text{ where } \delta = \frac{\left(\epsilon - 2^{-m-1}\right)^d}{4}$$

Proof idea. When restricting to n-tuples in K which are all in the image of H_m, the Hilbert canonization can be written as $H_m \circ \mathrm{sort} \circ H_m^{-1}$. Since sort (Proposition 3.3) is an isometry, the image of $\mathrm{sort} \circ H_m^{-1}$ can be covered efficiently. This cover can then be pushed forward to cover the image of $H_m \circ \mathrm{sort} \circ H_m^{-1}$, using the fact that H_m is Holder continuous. □

Theorem 4.3 shows that the covering number of the Hilbert canonization grows at worst polynomially in n, for fixed $\epsilon > 0$. This is in contrast with the exponential growth of the sorting canonization, proving a substantial gap between them.

Conclusion 3

Our analysis proves that the covering number of Hilbert canonizations is far superior to lexsort canonization, giving a theoretical justification for its superior performance in practice.

To obtain more intuition for the gap between our bounds for the covering numbers, in Table 4.1 we present bounds for the covering number of the quotient space, the Hilbert canonization and the sorting canonization, as well as the covering number of the hypercube $[0, 1]^{d\times n}$ which is known to be k^{nd} when $\epsilon = 1/(2k)$ (see proof in A.5). We take characteristic values encountered in point cloud learning scenarios: point dimensions are $d = 3$, a very modest value of $\epsilon = 1/6$, and n varying between 250 and 2000.

Table 1: *Comparison of covering number bounds as a function n between the quotient space, the Hilbert and Lexsort canonizations, and the hypercube*

Covering number vs. n	$\mathbf{n = 250}$	$\mathbf{n = 500}$	$\mathbf{n = 750}$	$\mathbf{n = 1000}$	$\mathbf{n = 2000}$
Quotient (upper bound)	2.1×10^{36}	7.4×10^{43}	2.2×10^{48}	3.5×10^{51}	2.0×10^{59}
Hilbert (upper bound)	5.3×10^{193}	7.9×10^{278}	5.0×10^{336}	5.0×10^{380}	4.4×10^{494}
Lexsort (lower bound)	1.1×10^{239}	4.0×10^{477}	1.4×10^{716}	5.2×10^{954}	9.2×10^{1908}
Hypercube (exact)	6.9×10^{357}	4.8×10^{715}	3.3×10^{1073}	2.3×10^{1431}	5.3×10^{2862}

We note that our bounds for the quotient space and Hilbert canonizations are upper bounds, while the Lexsort bound is a lower bound. Thus, the gap between the covering number of Lexsort and that of the other methods may be even larger than shown in the table. We also note that despite the conservative choice of $\epsilon = 1/6$, all covering numbers in the table are enormous. This picture may be over-pessimistic, as the 'true data distribution' will be supported on a 'small' set $K \subseteq [0, 1]^{d\times n}$ of 'realistic point clouds'. An experiment estimating the covering number on the Modelnet40 dataset will be shown later on in Table 2.

5 Experiments

We conduct several experiments to verify our theoretical insights and their manifestation in practical learning scenarios.

Covering Number on ModelNet In our first experiment, we considered the ModelNet [30] point cloud classification task. Our first goal was to see whether the covering number hierarchy between the four different methods considered in Table 4.1 is preserved when replacing the large domain $[0, 1]^{3\times n}$ with the 'ModelNet point-cloud domain' K. Namely, our interest is in estimating the covering number of K, its image under the lexsort and Hilbert canonizations, and the covering number of the quotient space $K/\mathbb{G}$.

We do not have access to the 'true point cloud' domain K, but only to the train and test sets. Thus, to estimate the covering number, we compute, for each test sample s, the minimum distance from s to all training samples (for computational efficiency, we consider only samples with the same label as s). We call this number q_s. We can then estimate the covering number by taking $\epsilon = \max_{s\in \text{Test}} q_s$, as this implies that the train set is an ϵ-cover of the test set. We call this measure the *maximal coverage*. We also consider a more robust measure, the *average coverage* $\text{mean}\{q_s|s \in \text{Test}\}$. The results, shown in Table 2, show that the expected covering number hierarchy is also preserved when considering real data.

Table 2: *Distance from each test sample to its nearest train sample for* $n = 256$ *on ModelNet10 and ModelNet40, reporting both mean and max over the test set. Lower is better.*

	Mean coverage ↓		**Max coverage** ↓	
Method	**ModelNet10**	**ModelNet40**	**ModelNet10**	**ModelNet40**
Regular distance	0.4545	0.3821	0.629	0.7304
Sort distance	0.3770	0.2988	0.5407	0.5550
Hilbert distance	0.2013	0.1955	0.5283	0.5490
Group distance	**0.0917**	**0.0957**	**0.2742**	**0.4880**

Classification on ModelNet Next, we consider whether the hierarchy in covering numbers between the different methods translates into a corresponding hierarchy in accuracy and generalization error. We choose a simple, purely non-invariant architecture: a Global MLP, and compare the performance of this network (i) without canonization, (ii) with lexsort canonization, and (iii) with Hilbert canonization (we cannot implement group averaging here since the group is very large). The results, shown in Table 3, reveal that the expected hierarchy is obtained, with Hilbert performing best both on ModelNet40 and ModelNet10, followed by the lexsort canonization and a plain MLP. The Generalization error column (defined as Test acc.-Train acc.) indicates that this performance gap is due to improved generalization.

Table 3: *On the ModelNet classification task, we compare vanilla MLP architectures in three configurations: (a) no canon. (b) lexsort canon. and (c) Hilbert canon. As predicted by our theory, the Hilbert canon. leads to better generalization, which translates into improved overall accuracy.*

	ModelNet10		ModelNet40	
Model	Test Acc ↑	Gen. error ↓	Test Acc ↑	Gen. error ↓
MLP (no canon.)	0.57	0.43	0.442	0.558
MLP (lexsort)	0.755	0.226	0.693	0.2556
MLP (Hilbert)	**0.7815**	**0.1167**	**0.74**	**0.0861**

To further verify the generality of our findings, we also trained the three alternatives on smaller samples from the ModelNet40 dataset. We find that the Hilbert $>$ lexsort $>$ no canon. hierarchy is preserved in all experiments, as shown in Table 8. We note that this experiment is only illustrative; the results attained by simple vanilla MLPs, even with Hilbert canonization, are far from state-of-the-art. In the context of our theory, this can also be explained through the fact that successful models for this task are typically permutation invariant, and hence their generalization is governed by the covering number of the quotient space, which is always better than the covering number of canonizations (Conclusion 1).

We next conduct some experiments to corroborate conclusion 1, which claims that, in terms of generalization, group averaging is preferable to canonizations, which are in turn preferable to ignoring invariance altogether.

Group averaging on PCA frames Our first experiment is still with the ModelNet40 classification task, but while the previous experiments investigates canonization w.r.t. permutations, and implicitly relies on the fact that ModelNet models are aligned w.r.t rotations, in this experiment we use the invariant Deepsets [32] to deal with permutation invariant, and will focus on rotation invariance: we will assume the ModelNet models rotation degree of freedom is initially unknown, and is fixed using the PCA axes as suggested in [22]. Assuming the point cloud singular values are pairwise distinct, this procedure determines the point cloud orientation uniquely up to a $\{-1, 1\}^3$ symmetry.

We consider four ways to deal with this symmetry: (i) Pure PCA, where the symmetry is ignored (ii) Skewness canonization, where the global sign of the x, y and z coordinates is chosen so that their third moment is positive (iii) group averaging and (iv) group augmentation where in each epoch a random group element is applied to the input. This last example can be seen as an approximation of group averaging. The results of this experiment, detailed in Table 4 and visualized in Figure 3, demonstrate a clear performance hierarchy. Group averaging achieves the highest overall accuracy, while group augmentation and skewness canonization yield comparable final results, with all three methods significantly outperforming the pure PCA baseline. Furthermore, as highlighted in Figure 3, the group augmentation approach suffers from a noticeably slower convergence rate compared to deterministic skewness canonization.

Group averaging for image classification Lastly, we check our approach on image classification. We take the rotated MNIST [15] dataset, and consider the action of the $p4$ group, which is the four-element group generated by 90 degree rotations. We consider 4 approaches: simple CNN, the canonization from [14] with random or (CN(p4 frozen)-CNN) or learned CN(p4)-CNN weights, and a group average over the 4 elements of the group. As shown in Table 5, the canonizations outperform the simple CNN, but are outperformed by group averaging, as predicted in Conclusion 1. We also see in the table that the accuracy is correlated with the covering numbers of the different methods.

Table 4: *Test accuracy on ModelNet40 with a DeepSets model with orientation determined by PCA, and sign ambiguity handled via averaging, sign augmentation, canonization, or Pure PCA (ignoring the ambiguity).*

Method	**Accuracy** ↑
Sign Averaging	**0.760** $\pm$ 0.0025
Sign Augmentation	0.742 $\pm$ 0.0022
Skewness Canonization	0.742 $\pm$ 0.0050
Pure PCA	0.729 $\pm$ 0.0062

Table 5: *Accuracy and mean coverage on Rotated-MNIST which in invariant with respect to the group $p4$ of 90 rotations. We consider a non-invariant CNN, two canonizations from [14], and group averaging.*

Model	**Test Acc.** ↑	**Coverage** ↓
CNN	94.8 ± 0.13	0.185
CN(p4)-CNN	96.47 ± 0.24	0.176
CN(p4 frozen)-CNN	95.3 ± 0.2	0.176
AvgCNN	**97.3** ± 0.001	**0.167**

6 Conclusion

In this paper, we establish a methodology for comparing different canonizations with group averaging and other equivariant models by studying their covering number. We show the covering number of canonizations is at best the covering number of the quotient space (for isometric canonizations), and at worst the covering number of the original domain (for certain discontinuous canonizations). These results suggest that group-averaged models are preferable when computationally feasible. We also show that the covering number of Hilbert canonizations for point clouds is substantially better than that of lexsort canonizations.

Limitations and Future Work A limitation of our analysis is that it relies on upper bounds on the generalization error (Proposition 2.2), which may not be tight. Nonetheless, our experiments indicate that lower covering numbers do correlate with better generalization and learning. Looking towards future developments, a central contribution of this work is the demonstration that a rigorous theoretical comparison between competing canonization methods can be established via their covering numbers. This framework complements the traditional focus on a model's ability to distinguish between complex symmetric examples, instead highlighting properties that arguably exert a more significant influence on the ultimate success of the learning process. In future work, we intend to apply our framework to other canonizations and extend it to enable a rigorous treatment of the generalization of frame averaging.

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A Proofs

In this section, we provide the proofs that are missing in the main text.

A.1 Generalization and Covering Numbers

Our generalization results are based on the results of [31]: a combination of Theorem 1 and Example 4 from [31] yields the following theorem:

Theorem A.1 ([31]). *If $\mathcal{Z}$ is compact w.r.t. metric ρ, and $l(h_S, \cdot)$ is Lipschitz continuous with Lipschitz constant $c(s)$, i.e.,*

$$|l(h_S, z_1) - l(h_S, z_2)| \leq c(S)\rho(z_1, z_2), \quad \forall z_1, z_2 \in \mathcal{Z},$$

and the training sample set S is generated by n IID draws from μ, then for any $\delta, \gamma > 0$, with probability at least $1 - \delta$ we have

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \leq c(S)\gamma + M\sqrt{\frac{2\mathcal{N}(\mathcal{Z}, \rho, \gamma/2)\ln 2 + 2\ln(1/\delta)}{n}}. \tag{5}$$

We now explain how Theorem 2.1 from the main text follows from Theorem A.1. We recall Theorem 2.1:

Theorem 2.1. *Let (X, ρ_X) and (Y, ρ_Y) be metric spaces and assume X is compact. Let $f : X \to Y$ be a c_f Lipschitz function, and let $\ell : Y \times Y \to [0, M]$ be a c_ℓ Lipschitz function. Then for any $\epsilon, \delta > 0$ and natural n, for a training sample set S generated by n IID draws from a distribution μ, with probability of at least $1 - \delta$, we have*

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(X, \rho_X, \epsilon)\ln 2 + 2\ln(1/\delta)}{n}}$$

where c_{h_S} denotes the Lipschitz constant of h_S.

Proof. We will apply Theorem A.1, where to apply it to to the setting discussed in Theorem 2.1, we set

$$z = (x, f(x)) \quad \mathcal{Z} = \{(x, f(x))| \quad x \in X\}, \quad \rho(z, z') = \rho_X(x, x') + \rho_Y(y, y'),$$

and

$$l(h_S, z) = \ell(h_S(x), f(x)).$$

The measure μ on X is pushed forward to a measure $\hat{\mu}$ on $\mathcal{Z}$ via the mapping $x \mapsto (x, f(x))$.

We note that l is Lipscitz with a Lipschitz constant of $c_\ell \max\{1, c_{h_S}\}$ as the following shows:

$$\begin{aligned}
|l(h_S, z) - l(h_S, z')| &= |\ell(h_S(x), y) - \ell(h_S(x'), y')| \\
&\leq c_\ell \left(\rho_Y(h_S(x), h_S(x')) + \rho_Y(y, y')\right) \\
&\leq c_\ell \left(c_{h_S}\rho_X(x, x') + \rho_Y(y, y')\right) \\
&\leq c_\ell \max\{1, c_{h_S}\} \left(\rho_X(x, x') + \rho_Y(y, y')\right) \\
&\leq c_\ell \max\{1, c_{h_S}\}\rho(z, z')
\end{aligned}$$

Next, we note that under the assumptions of Theorem 2.1, the covering number of $\mathcal{Z}$ is related to the covering number of X via

$$\forall \epsilon > 0, \quad \mathcal{N}\left(\mathcal{Z}, \rho, (1 + c_f)\epsilon\right) \leq \mathcal{N}(X, \rho_X, \epsilon) \tag{6}$$

since we have the upper bound

$$\rho\left((x, f(x)), (x', f(x'))\right) = \rho_X(x, x') + \rho_Y(f(x), f(x')) \leq \rho_X(x, x') + c_f\rho_X(x, x')$$

and therefore any cover of X by balls of radius ϵ is a cover of $\mathcal{Z}$ by balls of radius $(1 + c_f)\epsilon$.

Now to conclude the proof, choose any $\epsilon > 0$ and set $\gamma = 2(1 + c_f)\epsilon$. Since l is $c_\ell \max\{1, c_{h_S}\}$ Lipschitz, we have

$$\begin{aligned}|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| &\overset{(5)}{\leq} c_\ell \max\{1, c_{h_S}\}\gamma + M\sqrt{\frac{2\mathcal{N}(\mathcal{Z}, \rho, \gamma/2)\ln 2 + 2\ln(1/\delta)}{n}} \\ &= 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(\mathcal{Z}, \rho, (1 + c_f)\epsilon)\ln 2 + 2\ln(1/\delta)}{n}} \\ &\overset{(6)}{\leq} 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(X, \rho_X, \epsilon)\ln 2 + 2\ln(1/\delta)}{n}}\end{aligned}$$

which proves Theorem 2.1. □

Next, we prove proposition 2.2.

Proposition 2.2. *Let $(K, \rho, \mathbb{G})$ be a module. Let (Y, ρ_Y) be a metric space, and assume K is compact. Let $f : K \to Y$ be a c_f Lipschitz function* ***which is*** $\mathbb{G}$ ***invariant****, and let $\ell : Y \times Y \to [0, M]$ be a c_ℓ Lipschitz function , then for any $\epsilon, \delta > 0$ and natural n, for a training sample set S generated by n IID draws from the distribution μ, with probability of at least $1 - \delta$, we have*

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(h_S)\ln 2 + 2\ln(1/\delta)}{n}}$$

where

$$\mathcal{N}(h_S) = \begin{cases} \mathcal{N}(K, \rho, \epsilon) & \text{if } h_s \text{ is } c_{h_s} \text{ Lipschitz} \\ \mathcal{N}(c(K), \rho, \epsilon) & \text{if } h_s = \tilde{h}_s \circ c, \tilde{h}_s \text{ is } c_{h_s} \text{ Lipschitz, and } c \text{ is a canon.} \\ \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) & \text{if } h_s \text{ is } c_{h_s} \text{ Lipschitz and } \mathbb{G} \text{ invariant} \end{cases} \tag{1}$$

Proof. The case where h_S is Lipschitz (but not invariant) follows directly from Theorem 2.1.

Next, we consider the case that $h_S = \tilde{h}_S \circ c$ where $\tilde{h}_S$ is c_{h_S} lipschiz. We want to consider the domain to be $c(K)$, so consider the following distribution measure on $c(K)$:

$$\tilde{\mu}(A) = \mu(c^{-1}(A))$$

Note that

$$\forall k \in c(K), h_s(k) = \tilde{h}_s(k)$$

Next, sample n IID samples from K denoted by $x_1, x_2, \dots, x_n$ and note that $c(x_1), c(x_2), \dots, c(x_n)$ are an IID samples from $c(K)$ by $\tilde{\mu}$ and denote them by $s_1, s_2, \dots, s_n$. Note that

$$\mathcal{L}_{\text{emp}}(h_S) = \frac{1}{n}\sum_{i=1}^{n} \ell(h_S(x_i), f(x_i)) = \frac{1}{n}\sum_{i=1}^{n} \ell(\hat{h}_S(s_i), f(s_i)) = \mathcal{L}_{\text{emp}}(\hat{h}_S)$$

So, the empirical losses are the same, and next we show the expected losses:

$$\begin{aligned}\mathcal{L}_{exp}(\tilde{h}_S) &= \mathbb{E}_{x\sim\tilde{\mu}}\ell(\tilde{h}_S(x), f(x)) = \int_{c(K)} \ell(\tilde{h}_S(x), f(x))d\tilde{\mu} = \int_{c^{-1}(c(K))} \ell(\tilde{h}_S(c(x)), f(c(x))d\mu \\ &= \int_K \ell(h_S(x)), f(x)d\mu = \mathcal{L}_{exp}(h_S)\end{aligned}$$

Finally, we apply Theorem 2.1 on $\tilde{h}_S$ and as it's c_{h_S} lipschiz we obtain that with probablity at least δ holds that:

$$|\mathcal{L}_{exp}(\tilde{h}_S) - \mathcal{L}_{\text{emp}}(\tilde{h}_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(c(K), \rho_X, \epsilon)\ln 2 + 2\ln(1/\delta)}{n}}$$

And as $|\mathcal{L}_{exp}(\tilde{h}_S) - \mathcal{L}_{\text{emp}}(\tilde{h}_S)| = |\mathcal{L}_{exp}(h_S) - \mathcal{L}_{\text{emp}}(h_S)|$

We obtain that

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \leq 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(c(K), \rho_X, \epsilon)\ln 2 + 2\ln(1/\delta)}{n}}$$

In the second case, where h_S is c_{h_S} lipshiz and invariant, we want to consider the domain $K/\mathbb{G}$ and define $\pi : K \to K/\mathbb{G}$ by $\pi(k) = [k]$. Define the following distribution on $K/\mathbb{G}$ by:

$$\tilde{\mu}(A) = \mu(\pi^{-1}(A))$$

Define

$$\tilde{h}_S, \tilde{f} : k/\mathbb{G} \to \mathbb{R}$$

by

$$\tilde{h}_S([x]) = h_S(x), \tilde{f}([x]) = f(x)$$

Note that by invariance, those functions are well defined. Next, sample n IID samples from K denoted by $x_1, x_2, \dots, x_n$ and note that $\pi(x_1), \pi(x_2), \dots, \pi(x_n)$ are an IID samples from $K/\mathbb{G}$ by $\tilde{\mu}$ and denote them by $s_1, s_2, \dots, s_n$. Note that

$$\mathcal{L}_{\text{emp}}(h_S) = \frac{1}{n}\sum_{i=1}^{n} \ell(h_S(x_i), f(x_i)) = \frac{1}{n}\sum_{i=1}^{n} \ell(\hat{h}_S(s_i), \tilde{f}(s_i)) = \mathcal{L}_{\text{emp}}(\hat{h}_S)$$

So, the empirical losses are the same, and next we show the expected losses:

$$\begin{aligned}\mathcal{L}_{exp}(\tilde{h}_S) = \mathbb{E}_{x\sim\tilde{\mu}}\ell(\tilde{h}_S(x), \tilde{f}(x)) = \int_{K/\mathbb{G}} \ell(\tilde{h}_S(x), \tilde{f}(x))d\tilde{\mu} = \int_{\pi^{-1}(K/\mathbb{G})} \ell(\tilde{h}_S(\pi(x)), \tilde{f}(\pi(x)))d\mu = \\ \int_K \ell(\tilde{h}_S([x]), \tilde{f}([x]))d\mu = \\ \int_K \ell(h_S(x)), f(x)d\mu = \mathcal{L}_{exp}(h_S)\end{aligned}$$

Note that $\tilde{f}$ is c_f lipshiz function and $\tilde{h}_S$ is c_{h_S} lipshiz function. Finally, we apply Theorem 2.1 on $\tilde{h}_S$ and $\tilde{f}$ and as it's c_{h_S} lipschiz we obtain that with probablity at least δ holds that:

$$|\mathcal{L}_{exp}(\tilde{h}_S) - \mathcal{L}_{\text{emp}}(\tilde{h}_S)| \le 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon)\ln 2 + 2\ln(1/\delta)}{n}}$$

And as $|\mathcal{L}_{exp}(\tilde{h}_S) - \mathcal{L}_{\text{emp}}(\tilde{h}_S)| = |\mathcal{L}_{exp}(h_S) - \mathcal{L}_{\text{emp}}(h_S)|$

We obtain that

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \le 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(k/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon)\ln 2 + 2\ln(1/\delta)}{n}}$$

□

A.2 Proof for the equivariant case

Proposition A.2. *Let $(K, \rho, \mathbb{G})$ be a module. Let (Y, ρ_Y) be a metric space, and assume K is compact. Let $f : K \to Y$ be a c_f Lipschitz function* ***which is $\mathbb{G}$ equivariant****, and let $\ell : Y \times Y \to [0, M]$ be a c_ℓ Lipschitz function and $\mathbb{G}$ invariant, then for any $\epsilon, \delta > 0$ and natural n, for a training sample set S generated by n IID draws from the distribution μ, with probability of at least $1 - \delta$, we have*

$$|\mathcal{L}_{exp}(h_S) - \mathcal{L}_{\text{emp}}(h_S)| \le 2c_\ell \max\{1, c_{h_S}\}(1 + c_f)\epsilon + M\sqrt{\frac{2\mathcal{N}(h_S)\ln 2 + 2\ln(1/\delta)}{n}}$$

where

$$\mathcal{N}(h_S) = \begin{cases} \mathcal{N}(K, \rho, \epsilon) & \textit{if } h_s \textit{ is } c_{h_s} \textit{ Lipschitz} \\ \mathcal{N}(c(K), \rho, \epsilon) & \textit{if } h_s = \tilde{h}_s \circ c, \tilde{h}_s \textit{ is } c_{h_s} \textit{ Lipschitz, and } c \textit{ is a canon.} \\ \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) & \textit{if } h_s \textit{ is } c_{h_s} \textit{ Lipschitz and } \mathbb{G} \textit{ Equivariant} \end{cases} \tag{7}$$

Proof. We begin with the second case. Define $l(h_S, z) = \ell(h_S(x), f(x))$. We frist show it's $\mathbb{G}$ invariant w.r.t z:

$$l(h_S, g \cdot z) = \ell(h_S(g \cdot x), f(g \cdot x)) = \ell(g \cdot h_S(x), f(g \cdot x))$$

□

A.3 Proofs for Section 3

We restate and prove Proposition 2.4.

Proposition 2.4. *[See proof in A.3] Let $(K, \rho, \mathbb{G})$ be a module, and assume $\mathbb{G}$ is finite. Then*

$$\mathcal{N}(K, \rho, \epsilon) \leq |\mathbb{G}| \cdot \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon).$$

Proof. Let C be an ϵ-cover of $K/\mathbb{G}$ under $\rho_{\mathbb{G}}$, and define

$$\widehat{C} := \{g \cdot y \mid y \in C,\ g \in \mathbb{G}\}.$$

Then of course $|\widehat{C}| \leq |\mathbb{G}| \cdot |C|$.

We claim that $\widehat{C}$ is an ϵ-cover of K under d. Let $x \in K$. Since C covers $K/\mathbb{G}$, there exists $y \in C$ such that

$$\rho_{\mathbb{G}}([x], [y]) \leq \epsilon.$$

Because $\mathbb{G}$ is finite, the minimum in the definition of $\rho_{\mathbb{G}}$ is attained, so there exists $g \in \mathbb{G}$ such that

$$\rho(x, g \cdot y) \leq \epsilon.$$

By construction, $g \cdot y \in \widehat{C}$, hence $\widehat{C}$ covers K. Therefore $\mathcal{N}(K, \rho, \epsilon) \leq |\widehat{C}| \leq |\mathbb{G}| \cdot |C| = |\mathbb{G}| \cdot \mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon)$. □

We next prove Claim 3.2.

Claim 3.2. *[proof in A.3] Any isometric canonization $c : K \to K$ is (K, ϵ) optimal for all $\epsilon > 0$.*

Proof. As it always holds that

$$\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \leq \mathcal{N}(c(K), \rho, \epsilon)$$

we just need to prove the opposite direction.

Let C be a cover of $K/\mathbb{G}$ with $\mathcal{N}(K/\mathbb{G}, \rho, \epsilon)$ elements. Consider

$$\hat{C} := \{c(x) | x \in C\}$$

and we will prove that $\hat{C}$ is an ϵ cover of $c(K)$.

Let $y \in c(K)$, then by definition $\exists x \in K : y = c(x)$. By the definition of cover $\exists z \in C$ such that $\rho([x], [z]) \leq \epsilon$. But as $\rho_{\mathbb{G}}([x], [z]) = \rho(c(x), c(z))$ we obtain that $\rho(y, c(z)) \leq \epsilon$. So $\hat{C}$ is an ϵ cover of $c(K)$ as desired. □

Example 3.3. The following canonizations are isometries (see proof in A.3)

1. The canon. $c(x) = |x|$ with respect to the action of $\{-1, 1\}$ on $\mathbb{R}$ and the standard metric.
2. The canon. $\mathrm{sort} : \mathbb{R}^n \to \mathbb{R}^n$ with respect to the action of permutations and a metric ρ induced by a p norm.
3. The centralization canon. defined for point clouds in $\mathbb{R}^{d \times n}$ with respect to the action of translation by vectors in $\mathbb{R}^d$, with the metric ρ induced by the Frobenius norm.

Proof. The canonization $c(x) = |x|$ is an isometry as

$$\forall x, y \in K, \rho_{\mathbb{G}}([x], [y]) = \min\{|x - y|, |x + y|\} = ||x| - |y|| = |c(x) - c(y)|$$

The fact that sorting is an isometry with respect to the metric $\rho_{\mathbb{G}}$ induced from a p norm (which is the p-Wasserstein metric) is shown in Lemma 4.2 in [4].

consider the action of $\mathbb{R}^d$ on $\mathbb{R}^{d \times n}$ by translating all columns of a matrix in $\mathbb{R}^{d \times n}$ by a vector in $\mathbb{R}^d$. Let ρ be the metric induced from the Frobenius norm $\rho(X, Y) = \|X - Y\|_F$, and let c be the centralization mapping $c(X) = X - \frac{1}{n} X 1_{n \times n}$. This mapping is a canonization and an isometry. To see it is an isometry, note that we have for any canonization $\rho(c(X), c(Y)) \leq \rho_{\mathbb{G}}([X], [Y])$, and we get the converse inequality by noting that c is the orthogonal projection from the inner product space $\mathbb{R}^{d\times n}$ onto the subspace $\{X \in \mathbb{R}^{d\times n}| \quad X1_{n\times n} = 0\}$. Thus, let $g \in \mathbb{G}$ be a translation vector such that $\rho_{\mathbb{G}}([X],[Y]) = \rho(X, gY)$. Then

$$\rho_{\mathbb{G}}([X],[Y]) = \rho(X, gY) = \|X-gY\|_F \leq \|c(X)-c(gY)\|_F = \|c(X)-c(Y)\|_F = \rho(c(X), c(Y))$$

□

We restate and prove 3.4.

Proposition 3.4. *[proof in A.3] Let $(V, \rho, \mathbb{G})$ be a module, where $\mathbb{G}$ is a finite group. Let c be a canonization such that c has $|G|$ partial limits at a point $x \in V$, then for any small enough $\epsilon > 0$ there exists a $\mathbb{G}$ invariant set $K \subseteq V$ such that $\mathcal{N}(c(K), \rho, \epsilon) = \mathcal{N}(K, \rho, \epsilon) = |\mathbb{G}| \cdot \mathcal{N}(K/\mathbb{G}, \rho, \epsilon)$.*

Proof. L is a partial limit of c if there is a sequence $x_n \to x$ such that $c(x_n) \to L$. We claim that L must be in the orbit of x. Indeed, $c(x_n) = g_n x_n$ for appropriate $g_n \in G$. Since $\mathbb{G}$ is finite, by passing to a subsequence, we can assume that $g_n = g \in G$ does not depend on n. Since $x \mapsto gx$ is an isometry, we know that $gx_n \to gx = L$. In particular, $gx \neq x$ for every $g \neq e$. Choose some $\epsilon < min_{g\neq e}|x - g \cdot x|$, and define the G invariant set $K = K(\epsilon)$ by $K := \cup_{g\in G} B(g \cdot x, \epsilon)$. Note that $c(K)$ contains all partial limits of c at x, and therefore $gx \in c(K)$ for all $g \in G$. Since the distance between any two elements in the orbit of x is larger than ϵ, an ϵ cover of $c(K)$ must contain at least $|\mathbb{G}|$ elements. On the other hand, K itself is a union of $|\mathbb{G}|$ balls of radius ϵ, and therefore

$$\mathcal{N}(c(K), \rho, \epsilon) \leq \mathcal{N}(K, \rho, \epsilon) \leq |\mathbb{G}| \leq \mathcal{N}(c(K), \rho, \epsilon),$$

and therefore all these inequalities are equalities and the covering number is exactly $|\mathbb{G}|$.

Finally, note that $\rho(K/G, \rho_G, \epsilon) = 1$ as one ball of radius ϵ is enough for covering the quotient space, as all $|\mathbb{G}|$ balls are the same up to group action. □

A.4 Proofs for Section 4

For proving Lemma 4.2, we need the following lemma:

Lemma A.3. *Let $S := [m]^n$ all vectors with n elements, each between 1 and m. Define two vectors $v \equiv u \iff \exists \pi \in S_n : u = \pi \cdot v$. Then the number of equivalent classes is exactly $\binom{n+m-1}{n}$.*

Proof. Define $\forall i \in [m]$, c_i^u to be the number of times i appears in u.
Note that $u \equiv v \iff \forall i \in [m] : c_i^u = c_i^v$. So we want to ask how many solutions there are to the following system:

$$\sum_{i=1}^{m} c_i = n$$
$$\forall i, c_i \geq 0$$

By [24] we know the number of solutions to the equation is $\binom{n+m-1}{n}$. □

We now restate and prove 4.2.

Lemma 4.2. *Let n, d be natural numbers and $\epsilon \in (0, 1)$. Set $K = [0, 1]^{d\times n}$ and $\mathbb{G} = S_n$. Then the covering number of the quotient space satisfies*

$$\mathcal{N}(K/\mathbb{G}, \rho_{\mathbb{G}}, \epsilon) \leq \binom{n + k^d - 1}{n}, \text{ where } k = \left\lceil \frac{1}{2\epsilon} \right\rceil. \tag{4}$$

Proof. Set $k = \lceil 1/(2\epsilon) \rceil$. The set K can be covered by k^d hypercubes of diameter k and radius $1/(2k) \leq \epsilon$. Let C denote the collection of centers of these hypercubes, namely

$$C = \left\{ \left[\frac{2i_1 + 1}{2k}, \frac{2i_2 + 1}{2k}, \ldots, \frac{2i_n + 1}{2k} \right] | \quad i_j \in \{0, 1, \ldots, k - 1\}, 1 \leq j \leq n \right\}.$$

Applying the quotient map $\pi(c) = [c]$ we obtain points $\pi(C)$ in the quotient space which are an ϵ cover of $K/\mathbb{G}$. Note that points in C which are permutations of each other are mapped by π to the same orbit. Therefore, the cardinality of the cover $\pi(C)$ is the number of equivalence classes with respect to the relation on C defined by the group: $c_1 \equiv c_2 \iff \exists \pi \in S_n : c_1 = \pi(c_2)$. Using Lemma A.3, we know that the number of equivalence classes is

$$|\pi(C)| = \binom{n + k^d - 1}{n}$$

□

We now restate and prove 4.1.

Lemma 4.1. *[proof in A.4] Let $k, d, n \geq 2$ be natural numbers, and set $\epsilon = 1/(2k)$, $K = [0,1]^{d \times n}$ and $\mathbb{G} = S_n$. Then the covering number of the lexicographical sorting canonization satisfies*

$$\left(\frac{1}{2\epsilon}\right)^{(d-1)\cdot n+1} \leq \mathcal{N}(c_{\text{lex}}(K), \rho, \epsilon). \tag{3}$$

Proof. We consider the set $U \subset [0,1]^{d \times n}$ defined by

$$U = \{X \in [0,1]^{d\times n} | \quad X_{11} = X_{12} = \ldots = X_{1n}\}$$

The set U is not a subset of $c_{\text{lex}}(K)$, but it is a subset of its closure $\overline{c_{\text{lex}}(K)}$. This is because for every $X \in U$ we can apply an arbitrarily small perturbation of the first row of X so that it is ordered from small to large, and the X will be unaffected by the canonization. We then obtain

$$\mathcal{N}(c_{\text{lex}}(K), \rho, \epsilon) = \mathcal{N}(\overline{c_{\text{lex}}(K)}, \rho, \epsilon) \geq \mathcal{N}(U, \rho, \epsilon) \stackrel{(*)}{=} \left(\frac{1}{2\epsilon}\right)^{(d-1)\cdot n+1},$$

where $(*)$ follows from the fact that the hypercube $[0,1]^{(d-1)\cdot n+1}$ can be mapped isometrically (in the ∞ norm) to U via $f : [0,1]^{(d-1)\cdot n+1} \to U$ by repeating the first entry $n-1$ times, obtaining an $n \cdot d$ vector, and then transposing to be of shape $n \times d$, and from the fact that the covering number of the hypercube is $\left(\frac{1}{2\epsilon}\right)^{(d-1)\cdot n+1}$. □

A.5 Hilbert Canonization

In this appendix, we give a formal definition of Hilbert curves and a full proof of Theorem 4.3.

The Hilbert curve $H(x)$ is defined as a limit of the function $H_m : [0,1] \to [0,1]^d$, and these H_m are our main focus. We will now define these curves, loosely following the terminology from [11]. Firstly, for a given integer L, we consider a partition of the unit interval into 2^L disjoint intervals: the semi-closed intervals $I_L(\ell) = [\ell 2^{-L}, (\ell+1)2^{-L})$ for $\ell = 0, \ldots, 2^L - 2$, and the closed interval $I(2^L - 1) = [(2^L-1)2^{-L}, 1]$. We denote the collection of all these intervals by $\mathcal{I}_L$. For a natural $d > 1$, we also consider the partition of the unit hypercube $[0,1]^d$ into $2^{L\cdot d}$ disjoint hypercubes of the form $E^d_L(\vec{\ell}) = I(\ell_1) \times \ldots \times I(\ell_d)$. We denote the collection of these hypercubes by $\mathcal{E}^d_L$.

Next, for any given m, we define H_m to be a bijection from $\mathcal{I}_{d\cdot m}$ to $\mathcal{E}^d_m$, such that (i) adjacent intervals are mapped to adjacent hypercubes and (ii) if $I_{d\cdot(m+1)}(k') \subseteq I_{d\cdot m}(k)$ then $H_{m+1}\left(I_{d\cdot(m+1)}(k')\right) \subseteq H_m\left(I_{d\cdot m}(k)\right)$. An illustration of $H_m, m = 1,2,3$ for $d = 2$ is given in Figure 2. Each H_m maps a finite collection of intervals bijectively to a finite collection of hypercubes. This naturally induces a bijection from the grid defined by the intervals centroids, to the grid defined by the hypercube centroids. We denote these grids by $\mathbb{A}[d\cdot m] \subseteq [0,1]$ and $\mathbb{B}[d,m] \subseteq \mathbb{R}^d$, respectively.

The Hilbert curve H is obtained from H_m by a limiting procedure and shown to be Holder continuous [11]. Following similar ideas we show that each Holder function H_m is Holder continuous with constant that do not depend on m. This property will be crucial for the analysis of the covering number of the Hilbert canonization.

Lemma A.4. *For every natural m, d, we have that*

$$\forall x, y \in \mathbb{A}[d\cdot m], \quad \|H_m(x) - H_m(y)\|_\infty \leq 4|x-y|^{1/d}$$

Proof. Let $x \neq y$ be points in the one dimensional grid $\mathbb{A}[d \cdot m]$, and assume without loss of generality that $x < y$. Note that $y - x \geq \frac{1}{2^{d \cdot m}}$, since this is the minimal distance between points in the grid. On the other hand, $y - x \leq 1$. Therefore, there exists some k with $m - 1 \leq k \leq 0$, such that

$$2^{-d \cdot (k+1)} \leq |x - y| \leq 2^{-d \cdot k}. \tag{8}$$

It follows that x and y are in two adjacent intervals in the k-th level (or in the same interval), and by the nesting property of Hilbert's curve, we know that $H_m(x)$ and $H_m(y)$ are in two adjacent hypercubes (or in the same hypercube). Therefore, their distances is at most the sum of the diameters of those two cubes, so

$$\|H_m(x) - H_m(y)\|_\infty \leq 2 \cdot 2^{-k} \overset{(8)}{\leq} 4|x - y|^{1/d}$$

□

Hilbert Canonization As a first step, we define a canonization not on all of $[0,1]^{d \times n}$, but rather on the finite set of n tuples of grid elements $\mathbb{B}^n[d, m]$. This is done by mapping each tuple-element to the unit interval using H_m^{-1}, then sorting the resulting n-dimensional array, and mapping back to the d-dimensional grid using H_m. We denote this canonization by c_m. We can write $c_m = H_m \circ \mathrm{sort} \circ H_m^{-1}$, with the understanding that H_m and H_m^{-1} are applied elementwise to n-tuples of grid elements.

To extend H_m to all of $K = [0,1]^{d \times n}$, we employ the following procedure: For a given $X = (x_1, \ldots, x_n) \in K$, (i) **Rounding:** Each x_j resides in a unique hypercube in $\mathcal{E}_m^d$. Replace x_j with the centroid of this cube $y_j \in \mathbb{B}^n[d, m]$. (ii) **Sorting:** find the permutation $\tau \in S_n$ which sorts $\left(H_m^{-1}(y_1), \ldots, H_m^{-1}(y_n)\right)$ from small to large. (iii) **Canonizing** Apply the permutation τ to X. We note that if X contains several points in the same hypercube, the permutation τ is not uniquely defined. In this case, the order of points with the hypercube will be determined by lexicographical sorting.

We now restate and prove our theorem regarding the covering number of the Hilbert canonization:

Theorem 4.3. *Let $d, n, m \geq 2$ be natural numbers, let $\epsilon \in (0,1)$, and assume that $\epsilon > 2^{-m-1}$ set $K = [0,1]^{d \times n}$ and $\mathbb{G} = S_n$. Then the covering number of the Hilbert canonization c_m satisfies*

$$\mathcal{N}(c_m(K), \rho_\infty, \epsilon) \leq \binom{n + \lceil 1/(2\delta) \rceil - 1}{n}, \text{ where } \delta = \frac{\left(\epsilon - 2^{-m-1}\right)^d}{4}$$

Proof. We first begin by a reduction from covering K to covering the grid $\mathbb{B}^n[d, m]$ For convenience we abbreviate $\mathbb{B}^n := \mathbb{B}^n[d, m]$. We note that for any $X \in K$, the matrix Y obtained by the rounding procedure satisfies $\|X - Y\|_\infty \leq 2^{-m-1}$. This stays true also after applying the canonization, $\|c_m(X) - c_m(Y)\|_\infty \leq 2^{-m-1}$, since the canonization will apply the same permutation to both X and Y. It follows that

$$\mathcal{N}\left(c_m\left(K\right), \rho_\infty, \epsilon\right) \leq \mathcal{N}\left(c_m\left(\mathbb{B}^n\right), \rho_\infty, \epsilon - 2^{-m-1}\right).$$

It remains to bound the right-hand side:

$$\begin{aligned}
\mathcal{N}\left(c_m\left(\mathbb{B}^n\right), \rho_\infty, \epsilon - 2^{-m-1}\right) &= \mathcal{N}\left(H_m \circ \mathrm{sort} \circ H_m^{-1}\left(\mathbb{B}^n\right), \rho_\infty, \epsilon - 2^{-m-1}\right) \\
&\overset{(*)}{\leq} \mathcal{N}\left(\mathrm{sort} \circ H_m^{-1}\left(\mathbb{B}^n\right), \rho_\infty, \delta\right) \\
&\leq \mathcal{N}\left(\mathrm{sort}\left([0,1]^n\right), \rho_\infty, \delta\right) \\
&\overset{(**)}{\leq} \binom{n + \lceil 1/(2\delta) \rceil - 1}{n}
\end{aligned}$$

Where (*) follows from the Holder properties of H_m proven in Lemma A.4, and (**) follows from the fact that one-dimensional sorting is an isometry (Proposition 3.3) and the upper bound on the covering number of $[0,1]^n/\mathbb{G}$ (Lemma 4.2). □

Lemma A.5. *Be $K = [0,1]^{n \times d}$ and $\epsilon = \frac{1}{2 \cdot k} > 0$ such that $k \in \mathbb{N}$. Then $\mathcal{N}(K, d_\infty, \epsilon) = \frac{1}{(2\epsilon)^{nd}}$*

Proof. It's easy to see that each line we can cover using $\frac{1}{2\cdot\epsilon}$ ϵ sized lines, so using the product of size covering for $n \cdot d$ times we get a covering for K that uses $\frac{1}{(2\epsilon)^{nd}}$ balls. On the other hand, denote by F some other covering. Note that $K \subseteq \cup_{c\in F} B(c,\epsilon)$ So $1 = \mu(K) \leq \mu(\cup_{c\in F} B(c,\epsilon)) \leq \sum_{c\in F} \mu(B(c,\epsilon)) = |F| \cdot (2\epsilon)^{n\cdot d}$ so $\frac{1}{(2\cdot\epsilon)^{nd}} \leq |F|$. Thus, we got the lower bound and the upper bound as desired. □

B Additional Experiments

B.1 Laplacian Eigenvector Sign Ambiguity: Augmentation vs. Canonization on `ogbg-molpcba`

This appendix presents an additional empirical comparison between deterministic canonization and training-time random group augmentation, on a setting distinct from the main-text experiments and complementary to the PCA experiment of Section 4. The goal is to ask whether the empirical behaviour of these two strategies on a standard graph property prediction benchmark is qualitatively consistent with our Proposition 2.2 and the covering-number inequality (2); we do not claim a quantitative test of the inequality, since test AP on a finite-sample-trained GIN mixes optimisation, capacity and generalisation effects that the inequality does not isolate. The experiment is also consistent with the empirical observation cited in the Related Work that randomized SMILES can outperform canonized SMILES [1, 3].

Module. We study graph property prediction on `ogbg-molpcba` using a 5-layer Graph Isomorphism Network (GIN) with Laplacian positional encoding (LapPE). For each graph G with normalized Laplacian L_G, we compute the k smallest non-trivial eigenpairs $\{(\lambda_i, v_i)\}_{i=1}^k$ and concatenate the eigenvectors as a node-level positional encoding. The resulting LapPE is invariant to graph isomorphism only up to the eigenvector *ambiguity group*

$$\mathbb{G}_{\mathrm{LapPE}} = \prod_{i\,:\,m_i=1} \{-1,+1\} \times \prod_{i\,:\,m_i>1} O(m_i),$$

where m_i denotes the multiplicity of eigenvalue λ_i: a sign flip is admissible for each simple eigenvalue and an arbitrary orthogonal change of basis is admissible within each multiplicity-m_i block. On molecular graphs the spectrum is almost everywhere simple, so in practice $\mathbb{G}_{\mathrm{LapPE}}$ reduces to $\{-1,+1\}^k$ for almost all graphs. The natural questions are then (i) whether to canonize this ambiguity, and (ii) whether augmentation-based approximations of group averaging behave consistent with our framework.

Three methods compared. We compare three strategies for resolving $\mathbb{G}_{\mathrm{LapPE}}$, all implemented as faithful numpy ports of the reference torch code from the Laplacian canonization release of [19]. The first, `map` (Maximal Axis Projection), resolves a simple-eigenvalue sign by forming the rank-one projector $P_i = u_i u_i^\top$, grouping the coordinate indices j by the rounded column norms $\|P_i[:,j]\|$ (14-decimal rounding), and walking these groups from smallest to largest norm to find the first group whose indicator vector X (the sum of standard basis vectors over the group, plus a uniform shift) gives $\|P_i X\| \neq 0$; the canonical sign is then $\mathrm{sign}(u_i^\top X)$. For a multiplicity-$m_i$ block, `map` forms $P = U_i U_i^\top$, selects the m_i *largest*-norm coordinate groups, and builds the canonical orthonormal basis incrementally by projecting each group-indicator vector onto the residual orthogonal complement (with plain QR inside the complementary-space step). The second, `oap` (Orthogonalized Axis Projection), differs from `map` in two places: the sign step rounds the column norms to 6 decimals (coarser tie-grouping under near-degenerate norms), and the basis step replaces the largest-norm groupings of `map` by a hash-based grouping of columns and uses signed-QR in the complementary-space iteration. The third, `random_augmented`, flips the sign of each eigenvector with probability $1/2$ at every training-time forward pass, so the network sees a uniformly random element of $\{-1,+1\}^k$ at each step. This is a finite-group augmentation that approximates group averaging over $\mathbb{G}_{\mathrm{LapPE}}$ without summing over its 2^k elements explicitly. At evaluation time the `random_augmented` model is fed the deterministic eigsh-sign convention from the cache (no fresh random draws), yielding a single deterministic prediction per graph; this is the number reported in the `random_augmented` column of Table 6. Note that this is an *adversarial* test-time deployment of an averaging-trained model (it scores a single point of the orbit rather than the orbit mean); the

AUG-K column reports the alternative Reynolds-style sample of K orbit elements, and we discuss the gap between the two below.

Test-time averaging AUG-K. For `random_augmented` models we additionally evaluate a test-time *Reynolds-style* averaged metric. At evaluation, for each test graph we draw K independent group elements $g_1, \ldots, g_K \in \mathbb{G}_{\mathrm{LapPE}}$ (each g_j a sign-flip pattern on the simple-eigenvalue subspace and a Haar-random orthogonal matrix on each multiplicity-m block); we forward-pass the model on each sign-flipped LapPE; we average the predicted per-task probabilities; and we score the averaged predictions with the official OGB AP evaluator. We use $K=8$ at $k=3$ (which approximately covers $\{-1,+1\}^3$ under uniform i.i.d. sampling: 8 draws cover $\sim 65\%$ of the 8-element group on average) and $K=16$ at $k \in \{8, 16\}$ (sparse Monte-Carlo over $\{-1,+1\}^8$ and $\{-1,+1\}^{16}$ respectively). On `ogbg-molpcba` the spectrum is almost everywhere simple, so the multiplicity-block factor $\prod_i O(m_i)$ is degenerate in practice and the g_j samples reduce to sign-flip patterns. This is the empirical group-averaging operator of Section 2, restricted to a finite sample.

Hyperparameters and protocol. GIN, 5 layers, hidden dim $h \in \{16, 128, 512\}$, dropout 0.5, batch size 32, optimizer Adam with cosine learning-rate schedule and initial rate 10^{-3}, 3 seeds per cell ($\{0, 1, 2\}$), eigval-scaling off, eigenvector cache size 15. Best-validation-AP checkpoint selection. Software stack: PyTorch 2.10, CUDA 12.8, `torch_geometric`/`torch_scatter` per the environment file; OGB evaluator version per ogb's default `Evaluator(name='ogbg-molpcba')`. Per-arm seeding via `torch.manual_seed`, `numpy.random.seed` and `random.seed`; `cudnn.deterministic` not enforced (so results are not bit-reproducible across hardware). Standard deviations reported below are sample standard deviations (ddof$=$1). Code, raw `results.json` files, and reproduction commands are released at the project repository.

Compute budget. At $k \in \{3, 8\}$ all three arms are trained for 200 epochs with early-stop patience 15. At $k=16$, where validation AP for `random_augmented` continues to improve well past 200 epochs, all three arms are trained for 500 epochs with early stopping disabled (patience 999). Tables 6 and 7 therefore report a matched-compute comparison at every (k, h) cell. Both schedules are released in the project repository.

Results. Table 6 reports mean test AP at all nine (k, h) cells for the three methods, alongside the AUG-K test-time-averaged metric for `random_augmented`. Table 7 zooms in on the $k=16$ slice with sample standard deviations.

Table 6: Mean test AP on `ogbg-molpcba` at matched compute, 3 seeds per cell ($k \in \{3, 8\}$ at 200 epochs / patience 15, $k=16$ at 500 epochs / patience 999 for all three arms). Per-cell row-best margins are mostly within seed std and none survives a Bonferroni correction across the nine cells at $\alpha=0.05$ (see Table 7 for the $k=16$ Welch p-values); the bolding therefore marks the nominal row-best mean, not a significance claim. The AUG-K column reports test-time averaging over K ambiguity-group draws applied to the `random_augmented`-trained model (see text).

k	h	`map` ↑	`oap` ↑	`random_augmented` ↑	AUG-K
3	16	0.0626	0.0641	**0.0662**	0.0663 ($K=8$)
3	128	0.1704	0.1682	**0.1762**	0.1765 ($K=8$)
3	512	0.2397	**0.2431**	0.2395	0.2398 ($K=8$)
8	16	0.0622	0.0605	**0.0636**	0.0637 ($K=16$)
8	128	0.1705	0.1754	**0.1766**	0.1766 ($K=16$)
8	512	0.2344	0.2371	**0.2395**	0.2400 ($K=16$)
16	16	**0.0716**	0.0703	0.0691	0.0693 ($K=16$)
16	128	**0.1923**	0.1836	0.1916	0.1921 ($K=16$)
16	512	0.2468	0.2419	**0.2483**	0.2481 ($K=16$)

Table 7: Matched-compute $k = 16$ slice of Table 6: all three arms trained for 500 epochs with patience 999. Reported are mean $\pm$ sample standard deviation over 3 seeds, plus the two-sided Welch t-test p-value for the row-best mean against the row-second-best mean. None of the three row-best margins reaches $p < 0.05$, and after a Bonferroni correction across the nine cells in Table 6 ($\alpha = 0.05/9 \approx 0.0056$) none of the row-best margins anywhere in the appendix is significant.

h	`map`	`oap`	`random_augmented`	Welch p (best vs. 2nd)
16	$\mathbf{0.0716 \pm 0.0067}$	0.0703 ± 0.0070	0.0691 ± 0.0008	≈ 0.83
128	$\mathbf{0.1923 \pm 0.0005}$	0.1836 ± 0.0025	0.1916 ± 0.0009	≈ 0.32
512	0.2468 ± 0.0021	0.2419 ± 0.0028	$\mathbf{0.2483 \pm 0.0026}$	≈ 0.48

Test-time averaging Δ is small. Across all nine (k, h) cells the difference between the single-pass deterministic-eigsh evaluation of the `random_augmented` model and its AUG-K averaged evaluation (Table 6, right-most two columns) lies in $[-0.0002, +0.0005]$ AP, within the seed noise at every cell. We do not interpret this small Δ as conclusive evidence that the network is fully $\mathbb{G}_{\text{LapPE}}$-invariant: that would require a direct test (e.g. measuring per-graph output stability across adversarial group elements), which we do not run here. The operational reading is that, at the configurations tested, applying $K \in \{8, 16\}$ random group draws at test time does not measurably change AP relative to scoring on the cached deterministic eigsh-sign PE.

Reading the tables. Reading along the hidden-dim axis at fixed k in Table 6, the row-best AP grows monotonically with h at every k, and the row-best arm is most often `random_augmented` (six of nine cells), with `map` nominally row-best at $k = 16/h \in \{16, 128\}$ and `oap` nominally row-best at $k = 3/h = 512$. The matched-compute $k = 16$ slice in Table 7 resolves these margins against seed variance: the smallest sample std on each row is consistently the `random_augmented` std (0.0008, 0.0009, 0.0026), while `map` carries the largest std at small h (0.0067 at $h = 16$). Under a two-sided Welch t-test for unequal variances, the row-best versus row-second-best margins at $k = 16$ have p-values of 0.83, 0.32, and 0.48 at $h = 16, 128, 512$ respectively, none significant; this is the honest reading of the appendix at $n = 3$ seeds. At $k \in \{3, 8\}$ the row-best margins are similar in magnitude or smaller, with `random_augmented` winning five of six rows there but at margins of the same order as the seed std. We therefore present Table 6 as a nominal-rank summary, not as a benchmark claim.

Tie to the theory. On `ogbg-molpcba` at matched compute, with 3 seeds per cell, we did not detect a statistically significant difference between training-time random sign-flip augmentation and projection-based deterministic canonization at any of the nine (k, h) cells; this is qualitatively consistent with (2) of Proposition 2.2, which suggests augmentation can match canonization on tasks where the covering number of $\mathbb{G}$ is small, but it is not a positive test of the inequality, since AP gaps of order 10^{-3} are dominated by optimisation and seed noise rather than by the input-space covering complexity the inequality bounds. The AUG-K column moves AP by at most $+0.0005$ across the nine cells; this is consistent with the cached deterministic eigsh-sign PE already lying close to the orbit-mean the network was trained on, but is not itself a direct test of model invariance.

C Experimental Setup and Hyperparameter Details

C.1 Experimental Setup for the covering number experiment

We applied the following preprocessing for *each* sample: we first sample 256 points, then we shift it to the positive axis, and then divide by the maximum axis along x, y, z, obtaining all entries to be positive and between 0 and 1. The metric we considered on the point cloud space is

$$\rho_{\mathbb{G}}(X, Y) = \frac{1}{n} \sum_{i=1}^{n} \|X_i - Y_i\|_2.$$

The same metric is used for the two canonizations, and its quotient $\rho_{\mathbb{G}}$ is used for the quotient space.

Table 8: Test accuracy of the Global MLP under varying training set sizes. The performance gap between Hilbert and lexicographical sorting widens as data becomes scarcer, highlighting the regularizing effect of optimal canonization.

Training Samples	None ↑	Lexicographical ↑	Hilbert ↑
9840 (full)	0.433	0.700	**0.748**
4920	0.408	0.641	**0.714**
2460	0.343	0.573	**0.673**
1230	0.307	0.504	**0.601**

C.2 Experiment setup: ModelNet40 classification

In the ModelNet40 classification tasks, we sampled from each input CAD model point cloud, which consists of 1024 points, and applied an initial batched affine normalization to $[0,1]^3$. The data is then ordered using one of three strategies: no canonization (baseline), lexicographical sorting (i.e., sorting points sequentially by their x, then y, then z coordinates), or Hilbert curve sorting (using a space size of 2^{12}). Following ordering, the coordinates are projected into a higher-dimensional space using Random Fourier Features to capture high-frequency geometric details. The feature vector is then passed through a bottlenecked MLP consisting of residual blocks with dimensions $[256, 128, 64]$, yielding the final classification logits. The MLPs were trained for 100 epochs with a fixed batch size of 256, and were trained using AdamW.

To ensure a rigorous and fair comparison between the different ordering strategies, we conducted comprehensive grid sweeps to identify the optimal hyperparameters for the uncanonized baseline (`ply`), lexicographical sorting (`lex`), and Hilbert curve sorting (`hilbert`). The specific optimal hyperparameters extracted from our sweeps and utilized for the final evaluation are detailed in Table 9. The best hyperparameters for each model on ModlenNet40 are used for training the models on ModelNet10.

Table 9: Optimal hyperparameters identified via grid search for each ordering strategy.

Method	Learning Rate	Dropout	Weight Decay	Fourier Scale	Label Smoothing	Bands
No canon.	1.63×10^{-3}	0.3	0.05	1.0	0.1	2
Lexsort	6.41×10^{-4}	0.1	0.05	0.1	0.2	3
Hilbert	9.61×10^{-4}	0.3	0.01	0.1	0.2	3

Robustness Across Seeds To account for variance in weight initialization and optimization dynamics, we evaluated the final optimal configurations across 5 independent random seeds. The averaged test accuracy and standard deviations are reported in Table 10. The results demonstrate that the Hilbert curve canonization not only achieves the highest average accuracy but also exhibits the lowest variance across different initializations.

Table 10: Test accuracy across 5 independent random seeds using the optimal hyperparameters. Hilbert canonization yields both higher performance and greater stability.

Method	Average Accuracy ↑	Standard Deviation
No canon.	0.433	0.0249
Lexsort	0.700	0.01358
Hilbert	**0.748**	0.00944

C.3 Experimental Setup: Modelnet 40 Data scarcity experiment

To simulate data scarcity, we subsample the ModelNet40 training set using strides of 1, 2, 4, and 8, resulting in training subsets of 9840, 4920, 2460, and 1230 samples, respectively. Results are shown in Table 8.

C.4 Experimental Setup: PCA group averaging

We use a DeepSets architecture that is fully invariant to point order permutations but strictly non-invariant to the $\{-1, 1\}^3$ sign flips. We intentionally employ a lightweight, low-capacity version of this architecture. A highly over-parameterized model might achieve higher absolute accuracies, potentially masking the generalization gaps we aim to observe; by constraining the network capacity, we ensure that any performance differences are strictly attributable to the chosen alignment strategy. Each input point cloud consists of 1024 points from the ModelNet40 dataset. We compare four approaches: (1) **Pure PCA**, which leaves the sign ambiguity unresolved (acting as a baseline or potentially *poor* canonization due to eigensolver discontinuities); (2) **Skewness**, a deterministic canonization mapping $c : K \to K$ that resolves the ambiguity by orienting axes based on the third moment; (3) **Frame Averaging**, an explicitly $\mathbb{G}$-invariant model constructed by pooling the network's predictions across all 8 elements of the group using log-sum-exp; and (4) **Random Frame**, which applies elements of $\mathbb{G}$ as random data augmentation during training. All models are optimized using Adam for 1600 epochs and evaluated across 5 different random seeds.

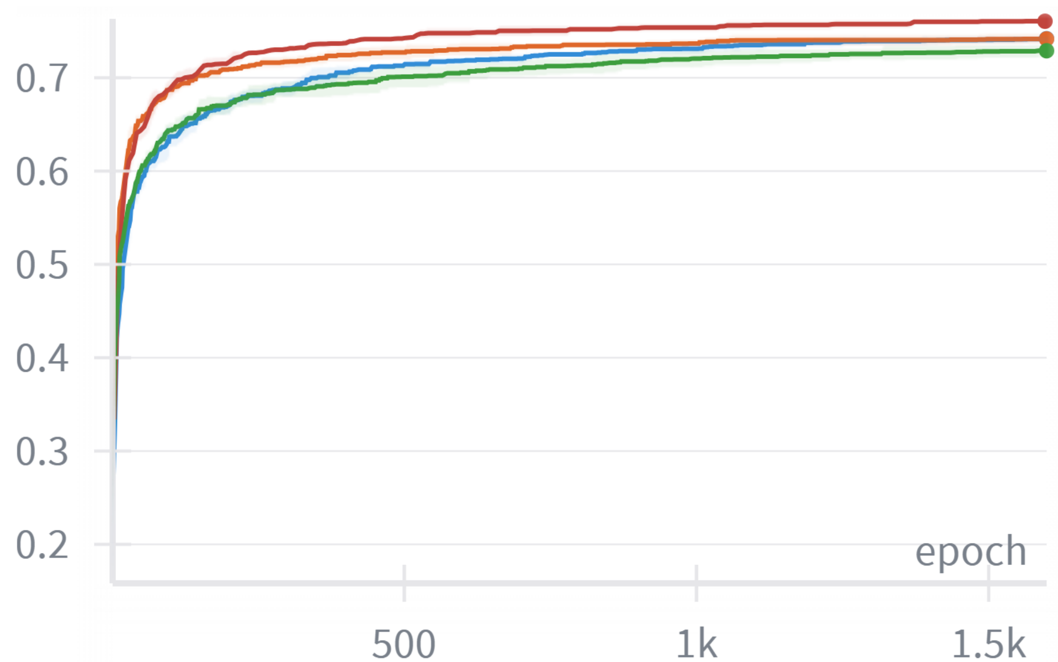

Figure 3: **Rotation Symmetry Accuracy.** The curves illustrate the comparative performance of the four approaches: Random Frame (blue), Frame Averaging (red), Skewness (orange), and the Pure PCA baseline (green). Frame Averaging achieves the strongest overall performance. While Random Frame and Skewness reach comparable final results—both significantly outperforming the Pure PCA baseline—the augmentation-based Random Frame method exhibits a noticeably slower convergence rate than deterministic canonization via Skewness.

C.5 Experimental Setup: Rotated MNIST

In the last set of the experiments, we compare the performance of several image rotation invariant models. We considered three models: simple CNN, CN(p4)-CNN, and AvgCNN. The backbone for all three models is the same and additional layers are added for CN(p4)-CNN, AvgCNN. In order to have the same setting, including the data split we rerun all models, including CN(p4)-CNN. We used learning rate of $1e^{-3}$ and a batch size of 256 and an AdamW optimizer. Each experiment is run for 5 different seeds, and mean and standard deviation are reported.

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